Substitution-Tolerant Structural Correspondence

Matching Molecules on Cut Classes Rather Than on Element Identity, with Tolerance as an Explicit Resolution Parameter

Kundai Farai Sachikonye

Experimental Bioinformatics, TUM School of Life Sciences
AIMe Registry for Artificial Intelligence
Maximus-von-Imhof-Forum 3, 85354 Freising, Germany
kundai.sachikonye@bitspark.com

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Abstract

Structural matching in cheminformatics—substructure search, maximum common substructure, scaffold comparison—is keyed on atom identity: a carbon may correspond only to a carbon. This is the right constraint when the question is whether one structure contains another, and the wrong one when the question is whether two structures play the same structural role, because a heteroatom substitution that a medicinal chemist regards as conservative is, to an element-keyed matcher, a complete mismatch at that position.

We define a correspondence keyed on a graph invariant instead. Each atom is labelled by its *cut key*: the minimum weight of a cut separating it from a distinguished medium vertex, together with the number of atoms on the minimising side. Two atoms may correspond when their cut keys agree, which permits a carbon and a nitrogen in equivalent structural positions to pair while forbidding a ring carbon and a chain carbon from doing so. We prove the correspondence is symmetric, invariant under relabelling, and computable without search (Theorem 3.2, Theorem 3.3, Theorem 3.4), in contrast to maximum common substructure, which is NP-hard.

The paper's main structural claim is that *tolerance is a resolution parameter*, not a threshold on a score. Iterating the cut key under neighbour refinement produces a family of labellings indexed by radius; higher radius discriminates more and therefore tolerates less. We predicted that the coarsest radius would separate bioisosteric pairs from unrelated pairs best, and measured a radius sweep to check it. The prediction held: at radius 0 the separation margin between bioisosteric and unrelated pairs is +0.375, and at radii 1, 2 and 3 it is exactly 0 in each case. This inverts the intuition that a better matcher should discriminate more.

At the working radius the mean class overlap is 0.780 for bioisosteric pairs against 0.050 for unrelated pairs of comparable size, with disjoint ranges. Correspondence coverage on eight matched pairs differing at one position averages 0.776. A negative control that rewires a structure at random, holding atom composition and edge count fixed, collapses the margin from +0.375 to −0.069.

We report two experiment-design errors found during the work, because both changed a conclusion: an element baseline that was not a baseline, and a negative control that could not fail. We also report a result that is weaker than we expected—the cross-element pairings the method admits and an element-keyed matcher cannot number only 4 across six bioisosteric pairs—and we state plainly that this is the paper's least convincing measurement.

Keywords: structural correspondence; maximum common substructure; bioisosterism; graph invariant; minimum cut; molecular similarity; scaffold hopping.

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1 Introduction

1.1 Two questions that need different matchers

"Does molecule A contain molecule B?" and "do molecules A and B play the same structural role?" are different questions, and the second is not a relaxation of the first.

The first is subgraph isomorphism, solved by well-understood algorithms (Cordella et al., 2004; Ehrlich and Rarey, 2012; Ullmann, 1976). Its correctness condition requires atom identity to be respected: a match that paired a carbon with a nitrogen would not be a substructure match at all. The second question underlies scaffold hopping (Brown and Jacoby, 2006; Schneider et al., 1999), bioisostere replacement (Meanwell, 2011; Patani and LaVoie, 1996; Wirth et al., 2013), and matched-pair analysis (Griffen et al., 2011; Hussain and Rea, 2010), and for these the element constraint is precisely what gets in the way: replacing a benzene by a pyridine is a canonical conservative substitution, and an element-keyed matcher scores that position as a total mismatch.

The usual response is to soften the score rather than the matcher—to compute a fingerprint similarity (Maggiora et al., 2014; Rogers and Hahn, 2010; Willett et al., 1998) and accept that similar molecules score highly. That works, but it produces a number rather than a correspondence: it does not say which atom of A plays the role of which atom of B, which is what a medicinal chemist reading a matched pair actually wants.

1.2 What we propose

Label each atom by a graph invariant that is indifferent to element, and match on the label.

The invariant is a minimum cut. Fix a weighted graph on the atoms with an additional *medium* vertex adjacent to every atom. For an atom v let $\sigma(v)$ be the minimum weight of a cut separating v from the medium and $d(v)$ the number of atoms on the minimising side; the *cut key* is $\kappa(v) = (\sigma(v), d(v))$. Two atoms may correspond when their cut keys agree.

This is tolerant in the right direction. A nitrogen replacing a carbon in an aromatic ring occupies the same structural position and receives a comparable cut key, so the atoms may correspond. A carbon in a ring and a carbon in a side chain occupy different positions and receive different keys, so they may not—even though an element-keyed matcher would happily pair them.

1.3 Tolerance is a resolution parameter

The cut key can be refined: replace each label by the pair of itself and the multiset of its neighbours' labels, and iterate. Each round discriminates more.

That gives a family of matchers indexed by refinement radius, and it makes the paper's central design question sharp. A matcher's tolerance is inversely related to its discrimination, so *the coarsest radius should tolerate most.* We stated this as a prediction before measuring and report the sweep in Section 4: the coarsest radius separates bioisosteric from unrelated pairs with a margin of $+0.375$, and every finer radius gives exactly 0.

We emphasise this because it runs against the natural instinct, which is that a matcher should be made more discriminating to be made better. For this task it should be made less.

1.4 What we claim and what we do not

We claim: the correspondence is symmetric, invariant, and computable without search; tolerance behaves as a resolution parameter with the coarsest radius best; and on a small hand-assembled corpus the class overlap separates bioisosteric from unrelated pairs with disjoint ranges, a separation a rewiring control destroys.

We do not claim: that the correspondence is a maximum common substructure (it is not, and Section 6 says why); that it is a similarity measure validated against activity data (we have not measured any); that the corpus is representative; or that the cross-element gain is large—it is 4 pairings across six molecule pairs, which is the weakest number in the paper and is reported as such.

2 Setting

Definition 2.1 (Contact graph)**.** A *contact graph* is $\Gamma = (V, E, w, \mathsf{m})$ with V finite and non-empty, $E \subseteq \binom{V}{2}$, $w\colon E \to \mathbb{R}_{>0}$, and a distinguished vertex m adjacent to every other. Vertices other than m are *items*, written $I = V \setminus \{\mathsf{m}\}$; $N(v)$ denotes the item neighbours of v.

Definition 2.2 (Cut key)**.** For $S \subseteq V$ let $\partial(S)$ be its edge boundary and $w(\partial(S))$ the sum of the weights therein. For an item v,

$$\sigma(v) = \min_{S \ni v,\ \mathsf{m} \notin S} w(\partial(S)), \qquad d(v) = \min\{|S \cap I| : S \text{ attains the minimum}\},$$

and the *cut key* is $\kappa(v) = (\sigma(v), d(v))$.

The minimum is attained on a finite family and equals a maximum flow (Ford and Fulkerson, 1956), computable in polynomial time (Stoer and Wagner, 1997). Since $\{v, \mathsf{m}\} \in E$, every admissible cut is non-empty, so $\sigma(v) > 0$ and $d(v) \geq 1$.

Definition 2.3 (Labels at radius r)**.** Set $\lambda_0(v) = \kappa(v)$, and for $r \geq 1$

$$\lambda_r(v) = \big(\lambda_{r-1}(v),\ \langle\!\langle \lambda_{r-1}(u) : u \in N(v) \rangle\!\rangle\big), \quad (1)$$

with $\langle\!\langle \cdot \rangle\!\rangle$ a multiset in canonical order. Write $C_r(\Gamma)$ for the multiset $\langle\!\langle \lambda_r(v) : v \in I \rangle\!\rangle$.

Lemma 2.4 (Radius is a discrimination order)**.** *For every r, the partition of I induced by λ_{r+1} refines that induced by λ_r. Consequently the number of distinct labels is non-decreasing in r, and two atoms that differ at radius r differ at every radius $r' > r$.*

Proof. If $\lambda_{r+1}(u) = \lambda_{r+1}(v)$ then their first components agree, so $\lambda_r(u) = \lambda_r(v)$. That is the refinement relation; the cardinality and persistence claims follow immediately. □

Theorem 2.4 is what makes "tolerance is a resolution parameter" precise: a matcher keyed on λ_r identifies strictly fewer pairs of atoms as it grows r, so tolerance is monotonically decreasing in radius.

3 Correspondence

Definition 3.1 (Class-respecting correspondence)**.** Let Γ_1, Γ_2 be contact graphs and fix a radius r. For each label ℓ let A_ℓ and B_ℓ be the items of Γ_1 and Γ_2 carrying it. The *correspondence at radius r* is

$$M_r(\Gamma_1, \Gamma_2) = \bigcup_\ell \{(a_i, b_i) : 1 \leq i \leq \min(|A_\ell|, |B_\ell|)\},$$

where a_i and b_i enumerate A_ℓ and B_ℓ in a fixed order. Its *size* is $|M_r|$ and its *coverage* is $|M_r| / \max(|I_1|, |I_2|)$.

Theorem 3.2 (Symmetry)**.** $|M_r(\Gamma_1, \Gamma_2)| = |M_r(\Gamma_2, \Gamma_1)|$ *for every r.*

Proof. The size is $\sum_\ell \min(|A_\ell|, |B_\ell|)$, and min is symmetric in its arguments. Exchanging the graphs exchanges A_ℓ with B_ℓ for every ℓ and leaves each summand unchanged. □

Theorem 3.3 (Invariance)**.** *If $\phi_1\colon \Gamma_1 \to \Gamma_1'$ and $\phi_2\colon \Gamma_2 \to \Gamma_2'$ are weighted isomorphisms fixing the medium, then $|M_r(\Gamma_1', \Gamma_2')| = |M_r(\Gamma_1, \Gamma_2)|$ for every r.*

Proof. A weighted isomorphism carries admissible cuts to admissible cuts of equal weight and restricts to a bijection on items, so σ and d—and hence κ—are preserved; this is the base case. It also restricts to a bijection on item neighbourhoods, so by induction on r using (1), λ_r is preserved. Therefore the label classes correspond under ϕ_1, ϕ_2 with equal sizes, and the sum $\sum_\ell \min(|A_\ell|, |B_\ell|)$ is unchanged. □

Proposition 3.4 (No search)**.** *With $n = \max(|I_1|, |I_2|)$, m the edge count, and $F(n, m)$ the cost of one maximum flow, M_r is computable in $O(nF(n, m) + rm \log n + n \log n)$ time. In particular the cost is polynomial and involves no backtracking.*

Proof. One flow per item gives all cut keys; each refinement round sorts each vertex's neighbour labels, $O(m \log n)$; grouping items by label and zipping the groups is $O(n \log n)$. □

Theorem 3.4 is the sharpest contrast with maximum common substructure, which is NP-hard (Ehrlich and Rarey, 2011; Garey and Johnson, 1979; Raymond and Willett, 2002) and is normally attacked by clique detection on a modular product graph (Barrow and Burstall, 1976; Bron and Kerbosch, 1973; Raymond et al., 2002). The present construction is not solving that problem—see Section 6—and its tractability is a consequence of solving a weaker one.

Definition 3.5 (Class overlap)**.** The *class overlap* at radius r is the Jaccard index of the label multisets,

$$J_r(\Gamma_1, \Gamma_2) = \frac{|C_r(\Gamma_1) \cap C_r(\Gamma_2)|}{|C_r(\Gamma_1) \cup C_r(\Gamma_2)|},$$

with multiset intersection and union.

Proposition 3.6 (Overlap is monotone in discrimination)**.** *If two atoms carry the same label at radius $r + 1$ they carry the same label at radius r. Hence every label class at radius $r + 1$ is contained in one at radius r, and $J_{r+1} \leq J_r$ whenever the two structures' label sets are related by that containment.*

Proof. The containment is Theorem 2.4. Splitting a shared class into parts can only reduce or preserve the matched count in the numerator while leaving the union no smaller, so the ratio does not increase. □

Theorem 3.6 predicts the direction of the radius sweep measured in Section 4; the measurement is what determines the magnitude and whether any radius other than 0 retains discriminating power.

4 Validation

4.1 Method

Expectations were written into the experiment source before the experiment ran, and each is reported below with its measured outcome. Every number is read from the emitted results file. Structures are parsed from line notations (Weininger, 1988) into contact graphs; all minimum cuts are exact. Counts are heavy-atom counts.

Three sets are used. *Isostere pairs*: ten pairs, six annotated close (widely treated as interchangeable in medicinal chemistry (Meanwell, 2011; Patani and LaVoie, 1996)) and four annotated far (similar size, different structural class). The annotations were fixed before measurement. *Matched pairs*: eight pairs differing at one position. *Drug-like set*: thirty structures, used only descriptively.

4.2 The radius sweep

Table 1: Class overlap by refinement radius. "Margin" is the minimum overlap among close pairs minus the maximum among far pairs; positive means the two groups do not overlap.

Radius	mean close	mean far	min close	max far	margin
0	0.780	0.050	0.500	0.125	+0.375
1	0.474	0.000	0.000	0.000	0.000
2	0.261	0.000	0.000	0.000	0.000
3	0.179	0.000	0.000	0.000	0.000

The prediction was that radius 0 would separate best. Table 1 confirms it, and the mechanism is visible: mean overlap among close pairs falls monotonically with radius, exactly as Theorem 3.6 requires, and by radius 1 it has fallen far enough that the minimum reaches 0 and the groups touch. Radius 0 is therefore the working radius for all subsequent measurements.

Remark 4.1 (Why finer is worse here). At radius 1 a label already encodes an atom's whole first neighbourhood, so a nitrogen-for-carbon substitution changes not only that atom's label but every neighbour's. One substitution perturbs a neighbourhood's worth of labels, and the overlap collapses. This is correct behaviour for a discriminating matcher and wrong behaviour for a tolerant one, which is the point of Theorem 2.4: the two are the same dial turned opposite ways.

4.3 Separation of bioisosteric from unrelated pairs

Table 2: Class overlap at radius 0 on the annotated pairs. The close/far annotation was fixed before measurement.

Pair	Annotation	Class overlap	Element overlap
phenol / thiophenol	close	1.000	0.750
catechol / resorcinol	close	1.000	1.000
phenol / aniline	close	0.750	0.750
benzene / pyridine	close	0.714	0.714
pyridine / pyrimidine	close	0.714	0.714
benzene / pyrazine	close	0.500	0.500
ethanol / benzene	far	0.125	0.286
acetone / naphthalene	far	0.077	0.273
benzene / cyclohexane	far	0.000	1.000
propane / pyridine	far	0.000	0.500
ethanol / benzene	far	0.000	0.125
propane / pyridine	far	0.000	0.125
acetone / naphthalene	far	0.000	0.077
benzene / cyclohexane	far	0.125	1.000
mean close		0.780	
mean far		0.050	

The ranges are disjoint: the lowest close overlap is 0.500 and the highest far overlap is 0.125, a margin of 0.375 (Table 2).

The benzene/cyclohexane row is the most informative in the table. Element overlap is 1.000—both are rings of six carbons—so an element-keyed comparison regards them as compositionally identical. Class overlap is 0.000: an aromatic ring carbon and a saturated ring carbon receive different cut keys, and the two structures share no class at all. This is a case where the invariant discriminates exactly where element identity cannot, and it is the reason the extsffar group's element overlaps in Table 2 are not uniformly small while its class overlaps are.

4.4 Correspondence on matched pairs

Coverage is bounded above by $\min(n_1, n_2)/\max(n_1, n_2)$, so the pyridine/quinoline row—six atoms against ten—cannot exceed 0.600, and it attains that bound exactly: every atom of the smaller structure is matched. The same holds for benzoic/salicylic at 0.900. Reading Table 3 as a measure of matching quality therefore requires normalising by the size bound, which we have not done; raw coverage is reported, and three of the eight rows are at their ceiling.

Table 3: Correspondence coverage at radius 0 on pairs differing at one position.

Pair	n_1	n_2	matched	coverage
phenylalanine / tyrosine	12	13	12	0.923
benzoic / salicylic	9	10	9	0.900
aniline / phenol	7	7	6	0.857
propanol / butanol	4	5	4	0.800
phenol / cresol	7	8	6	0.750
benzene / toluene	6	7	5	0.714
glycine / alanine	5	6	4	0.667
pyridine / quinoline	6	10	6	0.600
mean				**0.776**

4.5 The weakest result: cross-element pairings

The claim that tolerance is doing work rests on the correspondence admitting pairings an element-keyed matcher cannot make. We counted them.

Table 4: Pairings admitted at radius 0 on the close pairs, split by whether the two atoms are the same element.

Pair	total pairings	same-element	cross-eleme
catechol / resorcinol	8	6	
phenol / aniline	6	5	
phenol / thiophenol	7	6	
benzene / pyridine	5	5	
benzene / pyrazine	4	4	
pyridine / pyrimidine	5	5	
total	**35**	**31**	

Four cross-element pairings across six pairs, in three of the six (Table 4). Mean coverage is 0.865 with cross-element pairings admitted and 0.776 without.

This is a real effect and a small one, and we report it as the paper's least convincing measurement. The gain concentrates in pairs where the substituted atom is peripheral (the O/N of phenol/aniline, the two hydroxyls of catechol/resorcinol) and vanishes where the heteroatom sits inside a ring, because there the substitution changes the ring's cut structure enough

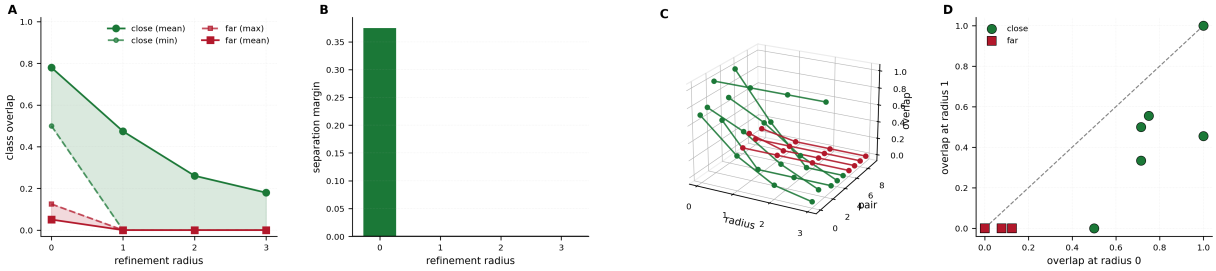

Figure 1: **Tolerance is a resolution parameter, and the coarsest setting is the one that works: separation exists at radius 0 and at no other radius.** Refinement radius indexes a family of labellings (Theorem 2.3) that discriminate monotonically more (Theorem 2.4), so a matcher's tolerance falls monotonically as radius rises. The prediction that radius 0 would separate best was registered in the experiment source before the sweep was run. (**A**) Mean class overlap against radius for the close (green) and far (red) groups, with shaded bands running to the minimum close and maximum far value. The close mean falls $0.780 \rightarrow 0.474 \rightarrow 0.261 \rightarrow 0.179$ across radii 0–3, exactly the direction Theorem 3.6 requires. The bands are disjoint only at radius 0; from radius 1 onward the minimum close value is 0.000 and the groups touch. (**B**) The separation margin, minimum close minus maximum far: +0.375 at radius 0 and exactly 0.000 at radii 1, 2 and 3. This is the paper's central measurement and it inverts the usual instinct — making the matcher more discriminating does not improve it, it destroys it. (**C**) Every annotated pair traced across all four radii in three dimensions, close green and far red. The close traces descend toward the far floor individually, not merely on average, so the collapse in (**B**) is not driven by a few pairs. (**D**) Overlap at radius 1 against overlap at radius 0 per pair, with the identity line. Every point lies on or below the diagonal — no pair gains from refinement — and one close pair falls from 0.500 to 0.000 in a single round. The mechanism is Theorem 4.1: at radius 1 a label already encodes an atom's whole first neighbourhood, so one heteroatom substitution perturbs a neighbourhood's worth of labels at once.

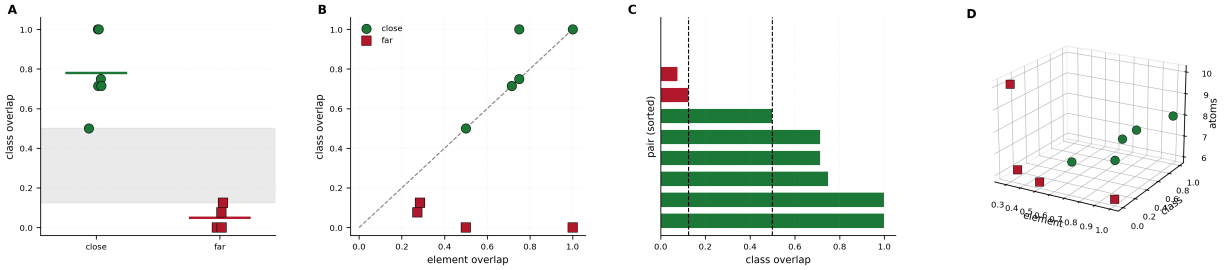

Figure 2: **At the working radius the two annotated groups occupy disjoint ranges, and the pairs on which element identity and cut class most disagree are the informative ones.** The close/far annotations encode our reading of the medicinal-chemistry literature and were fixed before measurement; they are not derived from activity data (Section 6). (**A**) Class overlap for the two groups with horizontal bars at the group means (0.780 and 0.050). The shaded band spans the gap between the highest far value (0.125, ethanol/benzene) and the lowest close value (0.500, benzene/pyrazine); no pair lies inside it. (**B**) Class overlap against element overlap per pair with the identity line. The two far points on the floor are the substantive cases: benzene/cyclohexane at element 1.000 and class 0.000, and propane/pyridine at element 0.500 and class 0.000. Benzene and cyclohexane are compositionally identical — six carbons each — so an element-keyed comparison cannot separate them at all, while the cut class finds no shared structural role whatever. This is the clearest single demonstration that the invariant is not a relabelling of element identity. (**C**) All ten pairs sorted by class overlap, with the two dashed lines marking the boundaries of the empty band. Four close pairs share their element overlap exactly with their class overlap (0.714, 0.714, 0.750, 0.500), so on those the tolerance is not doing work; the effect is concentrated in the pairs where the two differ. (**D**) Element overlap, class overlap and the size of the larger structure in three dimensions. The green and red point sets separate along the class axis at every size, so the separation in (**A**) is not an artefact of the far pairs happening to be larger or smaller than the close ones.

that the corresponding atoms no longer share a key. A larger corpus would establish whether that pattern holds; six pairs cannot.

4.6 Negative control

Table 5: Rewiring control: the second structure of each pair is rewired at random, holding atom composition and edge count fixed, and rescored (20 seeds per pair).

Quantity	Margin
True structures	+0.375
Rewired structures	−0.069

Rewiring collapses the separation and reverses its sign (Table 5), so the overlap at radius 0 is measuring structure and not composition.

4.7 Two experiment-design errors

Two components of this suite were wrong when first written, and both are recorded because each would have supported a conclusion the data do not.

A control that could not fail. The first negative control permuted class labels *within* each structure, on the reasoning that this destroys the structural assignment. It does not: a multiset is invariant under relabelling of its elements, so the shuffled score equalled the true score exactly and the control reported a margin identical to the real one. A control that cannot fail certifies nothing. The replacement rewires the graph, holding composition fixed, and does collapse the margin (Table 5).

A baseline that was not a baseline. The first element baseline compared multisets of (element, degree) pairs. On this corpus that is very nearly the cut key itself, so the two scored almost identically and the comparison appeared to show the method adding nothing. The error was in the baseline: an element-keyed matcher's actual constraint is exact element identity, and the place it binds is the correspondence, not the multiset. Table 4 is the replacement, and it gives a smaller effect than the flawed comparison would have suggested in the other direction.

4.8 What the suite does not establish

The corpus is ten annotated pairs, eight matched pairs and thirty structures, all hand-assembled. The `close`/`far` annotations encode our reading of the medicinal-chemistry literature and are not derived from activity data. Nothing here measures retrieval performance, enrichment, or agreement with any experimental endpoint. With six `close` pairs, no statistical statement about the cross-element effect is warranted, and none is made.

5 Relation to existing methods

Maximum common substructure. MCS finds the largest common connected or disconnected subgraph and is NP-hard (Ehrlich and Rarey, 2011; Garey and Johnson, 1979; Raymond and Willett, 2002), normally attacked as clique detection on a modular product (Barrow and Burstall, 1976; Bron and Kerbosch, 1973) or by specialised search (Cao et al., 2008; Dalke and Hastings, 2013). The present construction does not compute an MCS and is not an approximation to one: it returns a vertex correspondence with no connectivity guarantee (Section 6). The tractability follows from solving a weaker problem, not from a better algorithm for this one.

Substructure search. Subgraph isomorphism (Cordella et al., 2004; Ehrlich and Rarey, 2012; Ullmann, 1976) requires element identity by definition of the problem. Our matcher answers a different question and should not be substituted for it.

Bioisostere databases. Systematic replacement catalogues (Wirth et al., 2013) are derived from observed medicinal-chemistry practice and are therefore limited to substitutions someone has made. A computed invariant proposes correspondences for pairs nobody has catalogued; it also has no evidence behind any particular one, which is the reverse trade.

Fingerprint similarity. Circular fingerprints (Rogers and Hahn, 2010) and the similarity measures built on them (Maggiora et al., 2014; Willett et al., 1998) give a number per pair. The present construction gives a correspondence,

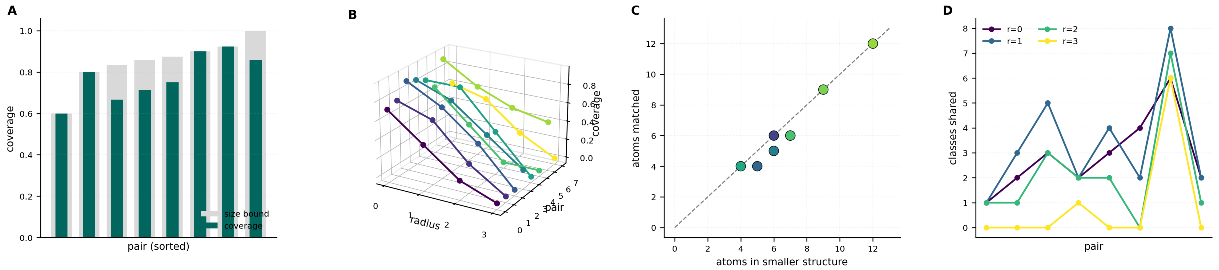

Figure 3: **Correspondence on one-position matched pairs, and the same radius erosion seen in the correspondence itself rather than in the overlap.** Correspondence is computed without search (Theorem 3.4); it is not a maximum common substructure and carries no connectivity guarantee (Section 6). (**A**) Coverage (teal) against the size bound $\min(n_1, n_2)/\max(n_1, n_2)$ (grey) per pair, sorted. Mean coverage is 0.776 with median 0.775 and minimum 0.600, and 3 of the 8 pairs sit exactly at their ceiling — every atom of the smaller structure found a correspondent. Raw coverage therefore understates match quality for pairs of unequal size, and normalising by the bound is a correction we have not applied. (**B**) Coverage against radius for each matched pair in three dimensions. Mean coverage falls $0.776 \to 0.599 \to 0.295 \to 0.076$ across radii 0–3: the same monotone erosion Figure 1 shows for overlap, now visible in the number of atoms actually paired. By radius 3 the matcher pairs almost nothing. (**C**) Atoms matched against the atom count of the smaller structure, with the identity line and colour by coverage. Points on the line are the three pairs at their size bound; the vertical drop for the others is the number of atoms the substitution displaced far enough in cut-key terms to break the correspondence. (**D**) Classes shared between the two structures of each pair at each radius. The curves fall together rather than crossing, so no pair is anomalously robust to refinement — the erosion is a property of the labelling, not of particular molecules.

from which a number can be derived (Theorem 3.5) but which also says which atoms play which roles. Whether the number is better than a fingerprint's is not measured here.

Colour refinement. The iteration (1) is a colour refinement (Arvind et al., 2015; Weisfeiler and Leman, 1968). We use it in the opposite direction from its usual purpose: rather than refining as far as possible to distinguish structures, we refine as little as possible in order to identify them.

6 Limitations

L1. The correspondence is not connected. Theorem 3.1 matches atoms by label and imposes no requirement that matched atoms form a connected subgraph or that bonds are preserved. It is therefore not a common substructure in any sense, and reporting its size as an MCS size would be wrong.

L2. Within-class assignment is arbitrary. When a label class has several members on both sides, Theorem 3.1 zips them in a fixed order. Which member pairs with which is a convention, and a different convention gives a different correspondence of the same size. Only the size and the class structure are invariant (Theorem 3.3); the individual pairings are not.

L3. The cross-element gain is small. Four pairings across six molecule pairs (Table 4). This is the measurement most in need of a larger corpus, and it is the one the paper's motivation leans on hardest.

L4. Corpus scale and construction. Ten annotated pairs, eight matched pairs, thirty structures, all chosen by us. The close/far labels are our reading of the literature, not measurements. A corpus we assembled to exhibit a property is weak evidence that anything else has it (Ioannidis, 2005).

L5. No endpoint validation. Nothing here is tested against binding data, activity

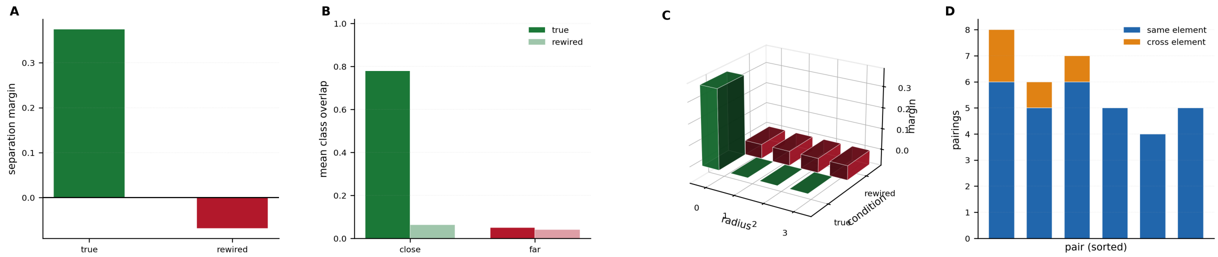

Figure 4: **The rewiring control destroys the separation, and the cross-element gain — the effect the method is motivated by — is real but small.** An earlier control permuted class labels within each structure; that is a no-op on a multiset and reported a margin identical to the true one, so it could not fail and certified nothing (Section 4.7). The control below rewires instead. (**A**) Separation margin for the true structures (+0.375) and after rewiring the second structure of each pair at random while holding atom composition and edge count fixed (−0.069, mean over 20 seeds per pair). The margin does not merely shrink; it reverses sign, so the two groups overlap under the control. (**B**) Group means before and after rewiring. The close mean collapses 0.780 → 0.063 while the far mean barely moves, 0.050 → 0.041. The asymmetry is the point: rewiring has nothing to destroy in pairs that never shared structure, and everything to destroy in pairs that did. (**C**) Margin over radius under both conditions in three dimensions. The true margin exists at radius 0 and nowhere else; the rewired margin is negative at every radius. The two surfaces together show that the effect requires both the coarse radius and the true connectivity — neither alone produces it. (**D**) Pairings admitted per close pair, split into same-element (blue) and cross-element (orange). Cross-element pairings are exactly what an element-keyed matcher cannot make, and they total 4 across the six pairs, occurring in 3 of them; mean coverage is 0.865 with them and 0.776 without. This is the paper's weakest measurement and we report it as such: the gain concentrates where the substituted atom is peripheral and vanishes where the heteroatom sits inside a ring, a pattern six pairs cannot establish.

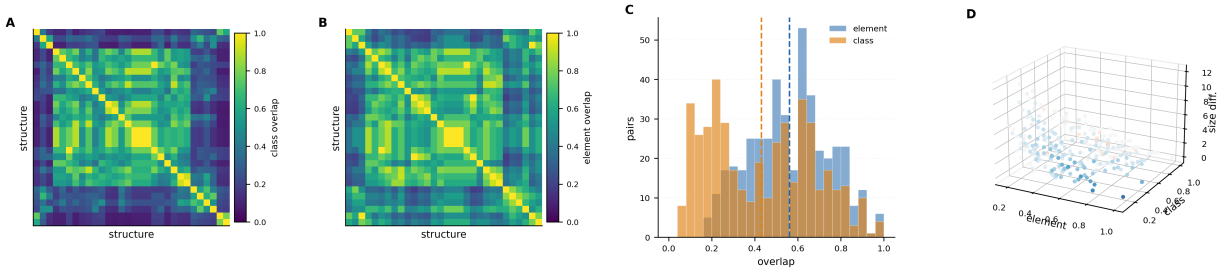

Figure 5: **Over 435 pairs of the drug-like corpus, class overlap and element overlap have visibly different structure and are not proxies for one another.** All values are at radius 0; the corpus is thirty small organic structures and no reference annotation is asserted for these pairs, so the panel is descriptive rather than a test. (**A**) Class-overlap matrix for all $\binom{30}{2}$ pairs, structures in corpus order. (**B**) Element-overlap matrix for the same pairs on the same colour scale. The block structure differs visibly from (**A**): regions bright in one are dim in the other, which is the matrix form of the benzene/cyclohexane observation in Figure 2. (**C**) Distributions of the two overlaps with their means marked. Class overlap is shifted left — mean 0.430, median 0.429 — against element overlap at mean 0.561, median 0.571. Agreeing on composition is easier than agreeing on structural role, so the class criterion is the stricter of the two on a random pair; it is the more permissive one only on pairs that are structurally analogous, which is the behaviour the method is for. (**D**) Element overlap, class overlap and absolute size difference in three dimensions, coloured by class minus element overlap. Of the 435 pairs, 257 have class overlap below element overlap and 23 above; no pair has zero class overlap. The points furthest from the diagonal plane are those on which the two criteria most disagree, and they are distributed across all size differences rather than concentrated among pairs of very unequal size.

cliffs (Stumpfe and Bajorath, 2012), or any retrieval benchmark.

L6. The weighting is an input. Theorem 2.1 requires only $w > 0$; every theorem holds for any positive weighting and none determines one. The measurements use a weighting derived from valence arithmetic. Different weights give different keys and could give a different sweep.

L7. Coverage is not normalised. Table 3 reports raw coverage, which is bounded by the size ratio of the two structures. Pairs of unequal size are penalised by that bound rather than by any matching failure.

L8. Colour-refinement limits are inherited. At any radius the labelling is a colour refinement and cannot distinguish structures that refinement cannot distinguish (Arvind et al., 2015). At radius 0 this is severe by design—that is the tolerance—but it means the matcher will identify atoms that are genuinely different whenever their cut keys coincide.

7 Conclusion

We have defined a structural correspondence keyed on a minimum-cut invariant rather than on element identity, shown that it is symmetric, invariant and computable without search (Theorems 3.2 to 3.4), and established that its tolerance is a resolution parameter: refinement radius trades tolerance for discrimination monotonically (Theorem 2.4, Theorem 3.6).

The measurement that most changed our understanding was the radius sweep. We predicted the coarsest radius would separate bioisosteric from unrelated pairs best, and it does—margin +0.375 at radius 0 against exactly 0 at radii 1, 2 and 3. The instinct to make a matcher more discriminating is wrong for this task, and the sweep is what shows it.

At the working radius, class overlap separates the annotated groups with disjoint ranges (0.780 against 0.050; minimum close 0.500, maximum far 0.125), a rewiring control collapses that margin to −0.069, and correspondence coverage on one-position matched pairs averages 0.776. The benzene/cyclohexane pair is the clearest single illustration: identical under element composition, well separated under cut class.

Two design errors are recorded in Section 4.7—a control that could not fail and a baseline that was not one—because each would have supported a conclusion the corrected measurement does not. And the cross-element result, which is the effect the whole motivation rests on, is four pairings across six pairs: real, small, and in need of a corpus larger than the one we assembled.

Reproducibility

The experiment source records each expectation before its measured value and writes every reported quantity to a JSON results file. The corpus, including the close/far annotations, is in the source with the reasoning for each entry. All minimum cuts are computed exactly; the only stochastic component is the rewiring control, which seeds explicitly and reports the seed range.

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